Private Cloud Architecture for Secure Drone Flight

A Self-Hosted, Secure and Real-Time Drone Data Management System

1. INTRODUCTION & OBJECTIVES

Problem Statement

Modern drones generate large amounts of sensitive data including live video, telemetry, and mission logs. Relying on public cloud services introduces privacy risks, internet dependency, and recurring operational costs. This project proposes a secure private cloud architecture that enables real-time monitoring, secure data storage, and authenticated communication between drones and operators.

Objectives

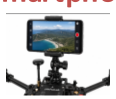
- ✓ Secure real-time drone communication
- ✓ Private cloud data storage
- ✓ Live video streaming and telemetry monitoring
- ✓ Strong authentication and role-based access control
- ✓ Scalable architecture for future multi-drone deployment

2. HARDWARE INTEGRATION

Drone frame

- Hexacopter Frame
- 6 Brushless Motors
- Pixhawk Flight Controller

Smartphone

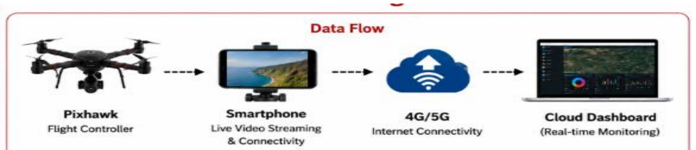


- Live Video Streaming
- Internet Connectivity
- Cloud Communication

Pixhawk Controller

- Stable Flight
- GPS Control
- Sensor Support

Hardware Integration



Key Hardware Benefits

- High Reliability
- Real-time Monitoring

3. SYSTEM ARCHITECTURE

DRONE (ONBOARD)

Camera Module 3

Pixhawk (Flight Controller)

- Capture Video
- Read Telemetry
- Authenticate Device
- Send Secure Data

Secure Connection (HTTPS / WebRTC / WebSockets!)

PRIVATE CLOUD (SELF-HOSTED)

- FastAPI Backend
- Authentication & Device Management
- Video Storage (Encrypted)
- Telemetry Database (PostgreSQL)
- User Management (RBAC)
- Security Logging & Monitoring
- PostgreSQL Database

DASHBOARD (USERS)

- Live Video Streaming
- Real-Time Telemetry
- Device Management
- Mission History
- Secure Access

SYSTEM WORKFLOW

1. Drone captures video and telemetry
2. Data is securely authenticated
3. Data is transmitted over encrypted channels
4. Private cloud stores and processes data
5. Users access data through dashboard
6. Real-time monitoring and mission management

4. SECURITY FEATURES

- Authentication
- Authorization
- Auto-Barrier (Intrusion Detection)
- HTTPS Encryption
- Secure Communication
- Real-time Validation
- Secure Storage

5. PUBLIC CLOUD vs PRIVATE CLOUD

Feature	Public Cloud (e.g., AWS)	Private Cloud (Our System)
Data Ownership	Third-Party	Complete Data Ownership
Security & Privacy	Variable	High (Built-in Security)
Scalability & Flexibility	High	Limited
Customized Cost	Medium	One-Time Setup
Deployment Control	Limited	100% Customizable
Server Maintenance	Vendor	Handled by Our System
Band Width / Latency	Variable	Low (Local Network)
Best For	General Purpose	For Secure Drone Operations
Internet Dependency	High	No (Offline Friendly)

5. FUTURE SCOPE

Raspberry Pi Deployment
Full onboard implementation for lightweight drones.

AI Integration
Object detection and anomaly detection using onboard AI.

Multiple Drone Support
Manage and monitor multiple drones simultaneously.

Advanced Security
End-to-end encryption, Mutual TLS, Secure Boot

Edge Analytics
Real-time processing and event detection at the edge.

Scalability & Expansion
Support for large-scale drone fleets and long-duration missions.

"Secure Data. Real-Time Insight. Complete Control."

All the submission of our project :

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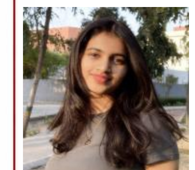

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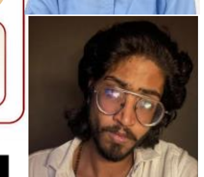

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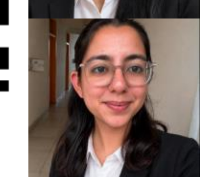

Hitanshi